Introduction to Module 9: Continuous Optimization

Design and Analysis of Algorithms
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Main Topics

- Analyzing convergence of iterative algorithms
- Numerical pitfalls / caveats
- Convexity
- Survey of algorithmic approaches for both unconstrained and constrained optimization.
- Tour of methods relevant for manipulating linear systems.
- Connections with sampling & diffusion-based generative AI.
- Optional enrichment lecture: multiplicative weight update methods and learning from experts.

Relevant Topics Previously Covered

- Nelder Mead / simplex search
- Population-based methods (e.g., genetic algorithms)
- Simulated annealing
- Duality
- Relaxation
- Divide and conquer algorithms for solving linear systems
- Iterative / alternating iterative refinement